SAGI Summer School 2025: Transfer Function Lab Overview

Sam Condon

July 24, 2025

Background

1. We have seen that for any LTI system, if our input is a summation of complex exponentials, our output is the same summation where each term is scaled by the system transfer function $H(\omega)$:

$$x(t) = \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \longrightarrow y(t) = \int_{-\infty}^{\infty} H(\omega) X(\omega) e^{i\omega t} d\omega \qquad (1)$$

2. In general, our transfer function is a complex number with both a real and imaginary component:

$$H(\omega) = \mathcal{R}\{H(\omega)\} + i\mathcal{I}\{H(\omega)\} \qquad (2)$$

3. Through Euler's formula $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$, this can be written as:

$$H(\omega) = |H(\omega)| e^{i\angle H(\omega)} \qquad (3)$$

where:

$$|H(\omega)| = \sqrt{\mathcal{R}\{H(\omega)\}^2 + \mathcal{I}\{H(\omega)\}^2}, \;\; \angle H(\omega) = \tan^{-1}\left(\frac{\mathcal{I}\{H(\omega)\}}{\mathcal{R}\{H(\omega)\}}\right) \qquad (4)$$

Background

1. When we are building an astronomical receiver, we need to know in detail the transfer function of our optics and all of our electronics.
2. For the electronics, we have circuit theory to help us design a circuit with the desired transfer function, but we need to perform a measurement to ensure that the circuit we design matches theory!
3. How can we measure the transfer function $H(\omega)$ of a circuit? In this lab we will design a system using a function generator and oscilloscope to perform this measurement!

Background

1. To understand the measurement setup, let's first consider the input and output pair of a single frequency signal at a frequency ω_0:

$$x(t) = e^{i(\omega_0 t + \phi_i)} \longrightarrow y(t) = H(\omega_0)e^{i\omega_0 t} = |H(\omega_0)|e^{i(\omega_0 t + \phi_i + \angle H(\omega_0))} \tag{5}$$

where ϕ_i is an arbitrary initial phase.

2. In the physical world, our signals correspond to the real part of complex exponentials, so our input/output pair becomes:

$$x(t) = \mathcal{R}\{e^{i(\omega_0 t + \phi_i)}\} = \cos(\omega_0 t + \phi_i) \tag{6}$$

$$y(t) = \mathcal{R}\{|H(\omega_0)|e^{i(\omega_0 t + \phi_i + \angle H(\omega_0))}\} = |H(\omega_0)|\cos(\omega_0 t + \phi_i + \angle H(\omega_0)) \tag{7}$$

Background

1. Now, consider taking the Fourier Transform of our input signal $x(t)$:

$$X(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \cos(\omega_0 t + \phi_i) e^{-i\omega t} dt \tag{8}$$

$$= \frac{1}{4\pi}\int_{-\infty}^{\infty} \left[e^{i(\omega_0 t + \phi_i)} + e^{-i(\omega_0 t + \phi_i)}\right] e^{-i\omega t} dt \tag{9}$$

$$= \frac{1}{4\pi}\int_{-\infty}^{\infty} \left(e^{i(\omega_0 - \omega)t} e^{i\phi_i} + e^{-i(\omega_0 + \omega)t} e^{-i\phi}\right) dt \tag{10}$$

$$= \frac{1}{2}\left[\delta(\omega - \omega_0)e^{i\phi_i} + \delta(\omega + \omega_0)e^{-i\phi_i}\right] \tag{11}$$

2. Doing the same for our output signal, we get:

$$Y(\omega) = \frac{|H(\omega_0)|}{2}\left[\delta(\omega - \omega_0)e^{i\phi_i} e^{i\angle H(\omega_0)} + \delta(\omega + \omega_0)e^{-i\phi_i} e^{-i\angle H(\omega_0)}\right] \tag{12}$$

Background

1. If we consider only positive frequency components ($\omega > 0$):

$$X(\omega)|_{\omega>0} = \frac{e^{i\phi_i}}{2}\delta(\omega - \omega_0) \tag{13}$$

$$Y(\omega)|_{\omega>0} = |H(\omega_0)|\frac{e^{i\phi_i}}{2}\delta(\omega - \omega_0)e^{i\angle H(\omega_0)} \tag{14}$$

2. Dividing the positive frequency output Fourier Transform by the positive frequency input Fourier Transform:

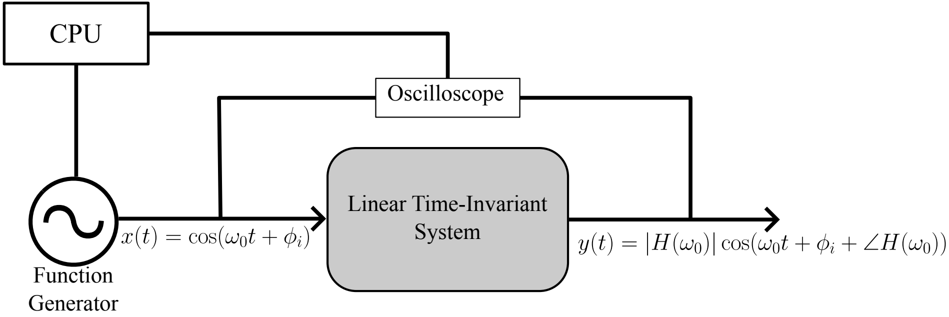
$$\frac{Y(\omega)|_{\omega>0}}{X(\omega)|_{\omega>0}} = |H(\omega_0)|e^{i\angle H(\omega_0)} = H(\omega_0) \tag{15}$$

   We see that this division recovers exactly the transfer function evaluated at $H(\omega_0)$

3. By sweeping the frequency of our input signal ω_0, we can measure the transfer function over several values!

Measurement Setup

❶ Our measurement setup is shown below:



Measurement Setup

1. We connect our computer to a function generator and oscilloscope.
2. The function generator creates the input signal at a single frequency ω_0. The oscilloscope records the input signal and output signal $x(t)$ and $y(t)$.
3. Using python, we Fourier Transform $x(t)$ and $y(t)$ and keep only the positive frequency components.
4. Finally, we obtain the complex value of the transfer function by dividing the positive frequency components of the output Fourier Transform by the positive frequency components of the input Fourier transform as outlined above.
5. We store the result $H(\omega_0)$, then repeat the process for multiple values of ω_0.
6. Since everything is automated on our computer, we can repeat the measurement for thousands of values of ω_0!